MUNNANGI VINAY

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PROFESSIONAL SUMMARY

Electrical and Electronics Engineering undergraduate with a CGPA of 8.68, possessing analytical, problem-solving, programming, and communication skills. Hands-on experience in technology-based engineering projects involving MATLAB, Simulink, and Python. Interested in business analysis, requirements understanding, customer-oriented problem solving, and technology-driven solutions. Seeking to contribute as a Business Analyst Trainee by analyzing business needs and supporting effective technology implementation.

EDUCATION

Velagapudi Ramakrishna Siddhartha Engineering College *2023 – 2027*
B.Tech in Electrical and Electronics Engineering | CGPA: 8.68/10

TECHNICAL SKILLS

Programming & Database: Python, SQL
Analytical & Simulation Tools: MATLAB, Simulink, MS Visio
Development Tools: STM32CubeIDE
Core Knowledge: Problem Solving, Analytical Thinking, Data Analysis, System Analysis, Requirements Understanding
Professional Skills: Communication, Teamwork, Adaptability, Quick Learning
Domain Knowledge: Electric Vehicles, Renewable Energy Systems, Power Electronics

ACADEMIC PROJECTS

High-Gain DC-DC Converter for DC Microgrid Applications

- Designed and simulated a high-gain DC-DC converter in MATLAB/Simulink by analyzing system requirements and evaluating voltage conversion performance.
- Implemented and compared five control strategies to evaluate their effectiveness under different operating conditions.
- Developed a Fuzzy Logic Controller for adaptive duty-cycle generation and analyzed system response across varying load and reference conditions.
- Conducted performance analysis including transient response, voltage regulation, switching frequency, and efficiency to support design decisions.

Electric Vehicle Fast Charging Station

- Designed and simulated AC-DC and DC-DC converter stages for an EV fast-charging system based on defined system requirements.
- Analyzed voltage regulation, current control, ripple characteristics, and transient response across multiple operating conditions.
- Validated system performance through simulation and tuned controller parameters to improve dynamic response and reliability.

CERTIFICATIONS

- NPTEL - Introduction to Internet of Things (IoT)
- NPTEL - The Joy of Computing Using Python
- NPTEL - Cloud Computing
- WISER - Quantum Fundamentals Program

PROFESSIONAL ACTIVITIES

- Student Member, IEEE
- Student Member, Institution of Engineers (India)
- Participated in an Electric Vehicle Workshop conducted by IIT Madras

AREAS OF INTEREST

• Business Analysis • Technology Consulting • Problem Solving • Data & Analytics • Product Solutions

• Digital Technologies • Customer-Oriented Solutions